The empirical recurrence for OEIS A199641, proved by a two-line transfer matrix

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31 August 2026

Abstract

OEIS A199641 counts arrays in which every cell is constrained by its neighbours on *both* sides, and which are moreover written in canonical form ("values introduced in row major order"). Neither feature is visible to a one-line transfer matrix: the cell condition reaches the row above and the row below, and the canonical-form clause is not local at all. Both are dealt with. What is counted is equality patterns, so $3!\,a(n) = \sum_i \binom{3}{i} D_{3-i} L_i(n)$ with D_m the derangement numbers and L_i the count over an i-letter alphabet; and each L_i is a walk count whose vertices are ordered pairs of consecutive rows. Relabelling-invariance makes the chain strongly lumpable, cutting 98 pairs down to $S = 23$ states. The entry's empirical recurrence of order 5, contributed by R. H. Hardin, then follows from a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 05A18, 15B36.

1 The sequence and the conjecture

OEIS A199641 is "Number of n X 2 0..2 arrays with values 0..2 introduced in row major order and each element equal to one or two horizontal and vertical neighbors.". It has offset 1 and begins

$$1, 3, 13, 60, 288, 1384, 6628, 31772, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 3*a(n-1) +7*a(n-2) +8*a(n-3) -8*a(n-5) for n>7.

As of the "Last modified" line on the live entry (May 16 2018 06:43:23, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 3\,a(n-1) + 7\,a(n-2) + 8\,a(n-3) - 8\,a(n-5) \tag{1}$$

for all $n > 7$.

2 The condition, and what the canonical-form clause counts

The arrays have 2 columns and $L = n$ rows. Write $x_{t,u}$ for the entry in row t and column u. with the neighbour offsets

$$\mathcal{N} = \{(-1,0),\ (1,0),\ (0,-1),\ (0,1)\}$$

written as (row shift, column shift), the requirement is

$$\#\big\{(d_1,d_2) \in \mathcal{N} :\ (t+d_1, u+d_2) \text{ lies in the array and its entry is } equaltox_{t,u}\big\} \ \in \{1,2\} \tag{2}$$

for every cell (t,u).

The condition is stated purely in equalities between entries, so it is unchanged by any relabelling of the values: it is a property of the *equality pattern* — the partition of the cells into classes of equal entries — and not of the values themselves. The clause "values introduced in row major order" selects exactly one array from each relabelling class, so with $K=3$ available values the entry counts admissible patterns with at most K classes.

Lemma 1. *Let $L_i(n)$ count the arrays over an alphabet of i letters that satisfy the cell condition, with no canonical-form clause, and $N_j(n)$ the admissible patterns with exactly j classes. Then $L_i = \sum_j N_j\, i(i-1)\cdots(i-j+1)$ and*

$$3! \sum_{j\le 3} N_j(n) \;=\; \sum_{i=0}^{3} \binom{3}{i} D_{3-i}\, L_i(n), \qquad D_m = m! \sum_{t=0}^{m} \frac{(-1)^t}{t!}.$$

Proof. A labelled array satisfying the condition is a pattern together with an injection of its classes into the alphabet, and a pattern with j classes admits $i(i-1)\cdots(i-j+1)$ of them; that is the first identity. Writing $i(i-1)\cdots(i-j+1) = j!\binom{i}{j}$ it reads $L_i = \sum_j (j!N_j)\binom{i}{j}$, so binomial inversion gives $j!N_j = \sum_i (-1)^{j-i}\binom{j}{i} L_i$. Summing over $j \le 3$ and exchanging the order of summation, the coefficient of L_i is $\sum_{t\le 3-i}(-1)^t/(i!\,t!)$, which multiplied by $3!$ is $\binom{3}{i}(3-i)!\sum_{t\le 3-i}(-1)^t/t! = \binom{3}{i}D_{3-i}$. □

3 Each L_i is a walk count on pairs of rows

Fix i and let $V_i = \{0,\dots,i-1\}^2$. Because (2) at a cell of row t involves rows $t-1$, t and $t+1$, a single row is not enough state. Take the ordered pairs $(p,c) \in V_i \times V_i$ as vertices and declare $(p,c) \to (c,x)$ an edge when every cell of the middle row c satisfies (2) with p above it and x below it. Let P_i be the adjacency matrix, $\mathbf{s}_i$ the indicator of the pairs (p,c) whose first row p satisfies the condition with no row above, and $\mathbf{e}_i$ the indicator of the pairs whose second row c satisfies it with no row below.

Lemma 2. *For $L \ge 2$, the arrays of L rows satisfying the condition are exactly the walks $(r^{(1)}, r^{(2)}) \to (r^{(2)}, r^{(3)}) \to \cdots \to (r^{(L-1)}, r^{(L)})$ of length $L-2$ that start at a vertex marked by $\mathbf{s}_i$ and end at one marked by $\mathbf{e}_i$, so $L_i = \mathbf{s}_i^\top P_i^{L-2}\mathbf{e}_i$.*

Proof. A walk of that form is exactly a sequence of rows $r^{(1)},\dots,r^{(L)}$, since consecutive vertices overlap in one row. In such a walk the condition at the cells of row t is tested once, at the step leaving $(r^{(t-1)}, r^{(t)})$, and with the correct neighbours $r^{(t-1)}$ and $r^{(t+1)}$; the cells of row 1 are tested by $\mathbf{s}_i$, which uses the condition with the row above absent, and those of row L by $\mathbf{e}_i$. So every cell is tested exactly once and with its true neighbourhood, which is the claim. □

Lemma 3. *Write $\pi(p,c)$ for the equality pattern of the pair. If $\pi(p,c) = \pi(p',c')$ then for every pattern Q the two vertices have the same number of successors of pattern Q, and $\mathbf{s}_i$, $\mathbf{e}_i$ take the same value at both. Hence the chain is strongly lumpable over π and L_i is computed on the quotient, whose states are the set partitions of the $2\cdot 2$ cells of a pair into at most i classes.*

Proof. Equal patterns means $(p',c') = (\sigma(p),\sigma(c))$ for a permutation σ of the alphabet. The edge condition and the two boundary conditions are statements about equalities between entries, so they are invariant under σ, and $x \mapsto \sigma(x)$ is a bijection between the successor sets preserving patterns. The number of pairs with a given pattern P is $e_P = i(i-1)\cdots(i-|P|+1)$, the number of injections of its classes into the alphabet, which supplies the weights on the quotient. □

Assembling the 3 lumped chains block-diagonally into N of size $S=23$, with the weights of Lemma 1 folded into the start vector $\mathbf{v}$ and the end-of-array indicator into $\mathbf{u}$,

$$3!\, a(n) \;=\; \mathbf{v}^\top N^{\,L-2}\mathbf{u}, \qquad L = n. \tag{3}$$

4 The criterion, and the computation

Let $q(t) = t^5 - \sum_i c_i t^{5-i}$ be the characteristic polynomial of (1) and put $G = N^1$, so that consecutive values of n advance the walk by 1 rows.

Lemma 4. *With $h_j = G^j N^o \mathbf{u}$ for the offset o that aligns the first admissible index,*

$$3!\Big(a(n) - \sum_i c_i\, a(n-i)\Big) \;=\; \mathbf{v}^\top G^{j-5}\left(h_5 - \sum_i c_i h_{5-i}\right)$$

for the corresponding j. Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for every $m \geq 0$, where $w = h_5 - \sum_i c_i h_{5-i}$; and it is enough to check $m = 0, 1, \ldots, S-1$.

Proof. Substitute (3) and factor. The scalar 3! is nonzero, so it does not affect vanishing. For the last clause, Cayley–Hamilton applied to G makes G^S an integer combination of $I, G, \ldots, G^{S-1}$, so each $\mathbf{v}^\top G^m w$ with $m \geq S$ is the same combination of the earlier values; if those vanish, so do all the rest. □

Theorem 5. *The recurrence (1) holds for every $n > 7$, which is the statement of Section 1.*

Proof. The lumped transition counts were built by taking one representative pair for each pattern, enumerating every row x over the alphabet, testing (2) on the middle row, and sorting the results by the pattern of the new pair. The vector w was formed by 5 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \ldots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 7+1$ onwards, and the run was continued until $S+1$ consecutive zeros had been seen, which by Lemma 4 settles every larger m. □

5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry's DATA: the right-hand side of (3), divided by 3!, was evaluated at every one of the 23 published indices and agrees exactly, the quotient coming out an integer each time. This tests the modelling and not only the arithmetic — the inversion of Lemma 1, the reading of (2), the treatment of the first and last rows, and the alignment of the entry's index with $L = n$ would each show up as a mismatch.

Second, the lumped chain was checked against the unlumped one on the small shapes of this family: the walk counts over the full alphabet agree with $\sum_P e_P f_m(P)$ term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 4 was carried to the full Cayley–Hamilton bound rather than to a sampled prefix.

References

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